Statistical Mechanics: Multiplicity, Entropy, and Constrained Equilibrium

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Lecture notes organized by LazyingArt LLC with Video2Book

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Chapter 1

Multiplicity, Entropy, and Constrained Equilibrium

Overview. A probability distribution becomes a thermodynamic object once its states carry energies. We first connect entropy increase to equilibration and use $dE = T\,dS$ to derive the direction of heat flow. We then replace one system by a large collection of replicas, count the arrangements associated with each set of occupation numbers, and show that the logarithm of this multiplicity is N times the entropy. Equilibrium is therefore a constrained maximum-entropy problem. Lagrange multipliers provide the method needed to solve it.

1.1 The Three Basic Quantities

Let i label the discrete states available to a system. For the moment, discreteness is useful because it lets us count states without introducing phase-space measures. If the probability of state i is P_i, then

$$\sum_i P_i = 1. \tag{1.1}$$

If the energy of that state is E_i, the mean energy is

$$\overline{E} = \sum_i P_i E_i. \tag{1.2}$$

The word *mean* matters. The system need not possess the value $\overline{E}$ in any one microscopic state; it is the probability-weighted average over the possible values.

> **Question for the class.** Is there a useful distinction here between probability and statistics?
>
> **Response.** Probability supplies a model for possible outcomes, while statistics usually begins with observed data and asks what can be inferred from it. The present calculation uses probability distributions, although the same averages naturally appear in statistical data. The informal exchange in class treats the words lightly because that distinction does not affect the physics that follows. [1]

[1] Lecture timestamp: 00:01:39–00:02:21.

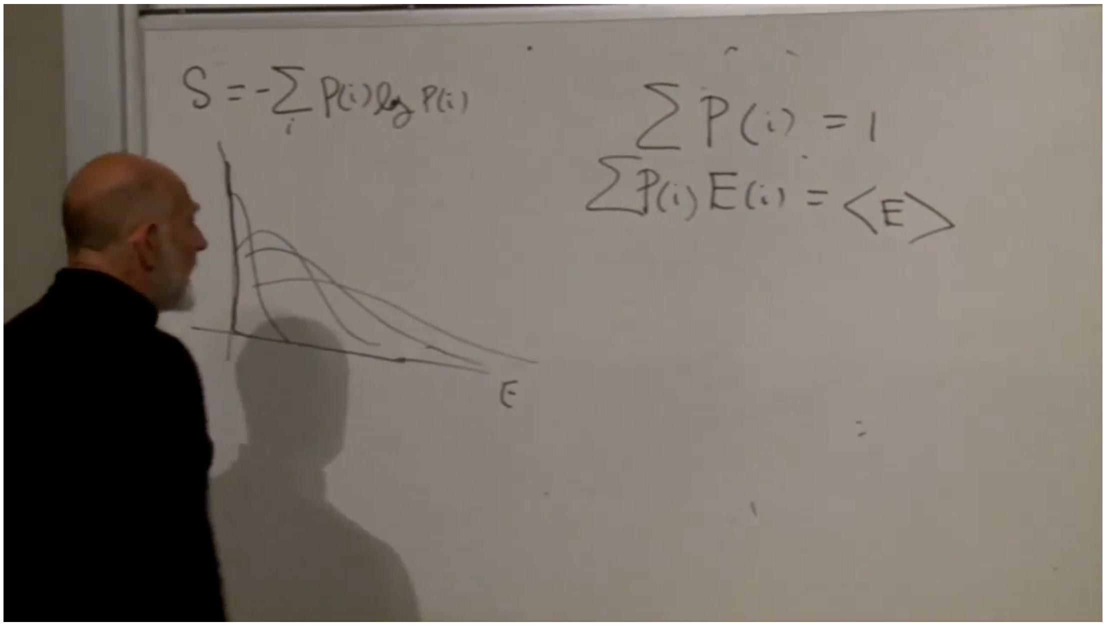

Figure 1.1: The opening board collects the entropy, normalization, and mean-energy formulas together with a qualitative family of equilibrium distributions.

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The entropy of the distribution is

$$S[P] = -\sum_i P_i \log P_i. \tag{1.3}$$

This expression is nonnegative. Every nonzero probability obeys $0 < P_i \leq 1$, so $\log P_i \leq 0$, and the conventional limiting value $P_i \log P_i \to 0$ applies when $P_i \to 0$.

1.2 Equilibrium as a Family of Distributions

The previous lecture left us with a one-parameter family of equilibrium distributions. One may label that family by temperature, but for now it is more useful to label it by the mean energy $\overline{E}$. At the lowest possible energy the distribution is concentrated on a ground state:

$$P_i = \delta_{i0}.$$

There is then no uncertainty about the state, and $S = 0$. As the equilibrium distribution spreads over more energy levels, both its mean energy and its entropy rise.

This is a statement about the thermal family, not about every distribution that can be invented. An arbitrary family could shift toward larger energies while becoming narrower. The equilibrium distributions to be derived do not behave that way in the ordinary positive-temperature regime. Along that family,

$$\frac{dS}{d\overline{E}} > 0. \tag{1.4}$$

Systems with bounded spectra can also have branches on which this derivative is negative. Those negative-temperature regimes are real, but they are set aside during the present argument.

Thermal equilibrium means more than microscopic stillness. Molecules continue to move, collide, and exchange energy. What has become stationary is the macroscopic description: the distribution no longer exhibits a net drift, and separated portions of the system no longer sustain a net energy flow.

1.3 The Thermodynamic Laws in Statistical Language

The first law is energy conservation. The second law says that the thermodynamic entropy of an isolated macroscopic system does not decrease:

$$\Delta S_{\text{tot}} \geq 0. \tag{1.5}$$

An equality describes equilibrium or an ideal reversible change; a spontaneous relaxation ordinarily gives a strict increase.

In the probabilistic picture, relaxation tends to spread weight over the macroscopic alternatives that we still distinguish. This statement requires the qualification emphasized in the lecture: we are *coarse-graining*. Exact Hamiltonian evolution preserves fine-grained phase-space entropy. Macroscopic entropy grows because we cease to track microscopic correlations and details. Losing such information broadens the reduced description. A spontaneous narrowing would instead mean that we had learned microscopic information without performing a measurement.

The zeroth law asserts that systems can reach thermal equilibrium and that equilibrium is transitive. If A is in equilibrium with B, and B with C, then A is in equilibrium with C. Temperature supplies the number that makes this transitivity transparent:

$$T_A = T_B, \qquad T_B = T_C \quad \Longrightarrow \quad T_A = T_C.$$

Out of equilibrium, energy flows from the subsystem with larger temperature to the subsystem with smaller temperature. At equal temperatures the net flow ceases. We now derive that operational statement from the entropy definition of temperature.

1.4 Why Energy Flows from Hot to Cold

Consider two weakly coupled subsystems A and B. Each is internally equilibrated, but their temperatures differ. Choose the labels so that

$$T_B > T_A > 0. \tag{1.6}$$

During a small exchange, conservation of the combined energy gives

$$dE_A + dE_B = 0. \tag{1.7}$$

The second law gives

$$dS_A + dS_B \geq 0. \tag{1.8}$$

For each internally equilibrated subsystem, temperature is defined by

$$dE_X = T_X \, dS_X, \qquad X = A, B. \tag{1.9}$$

Substitution into (1.7) yields

$$T_A\,dS_A + T_B\,dS_B = 0, \tag{1.10}$$

and hence

$$dS_B = -\frac{T_A}{T_B}\,dS_A. \tag{1.11}$$

Putting this result into (1.8),

$$dS_{\text{tot}} = dS_A - \frac{T_A}{T_B}dS_A \tag{1.12}$$

$$= \frac{T_B - T_A}{T_B}\,dS_A \geq 0. \tag{1.13}$$

Because $T_B - T_A > 0$ and $T_B > 0$, a spontaneous exchange requires

$$dS_A > 0.$$

Equation (1.9) then gives

$$dE_A = T_A\,dS_A > 0, \qquad dE_B = -dE_A < 0. \tag{1.14}$$

Energy has flowed from B, the hotter subsystem, to A, the colder one. Here heat is simply energy in transit. The flow continues until $T_A = T_B$, when the first-order entropy change no longer favors either direction.

The same result can be displayed in one line:

$$dS_{\text{tot}} = dE_A\left(\frac{1}{T_A} - \frac{1}{T_B}\right) \geq 0. \tag{1.15}$$

The sign of the inverse-temperature difference fixes the sign of dE_A. This compact form will later fit naturally with the canonical ensemble.

> **Question from the audience.** Does this argument assume that A and B are each already in internal equilibrium?
>
> **Response.** Yes. Assigning a single T_A or T_B makes that assumption. If a subsystem is not internally equilibrated, divide it into smaller regions and apply the argument locally wherever a temperature can be defined. The global relaxation is then assembled from exchanges among those regions. [2]

1.5 What Counts as a State?

Before asking for the equilibrium probabilities, we must know what the label i means. A microscopic state is the most complete specification of the system allowed by the underlying laws of physics. In classical mechanics, one pictures the positions and momenta of all particles. In quantum mechanics, a state is instead a complete quantum state; the uncertainty principle prevents a simultaneous sharp specification of every classical position and momentum.

[2] Lecture timestamp: 00:24:38–00:25:22.

For a quantum system in a finite box, the relevant state counting can be discrete. Classical statistical mechanics emerges in an appropriate limit, where sums over densely spaced quantum states become phase-space integrals. That limiting process also introduces the phase-space cell needed to make continuous entropy dimensionless; it is not obtained merely by replacing a summation sign with an integral sign.

> **Question from the audience.** For a gas, should a state be imagined as the position and momentum of every molecule?
>
> **Response.** That is the correct classical picture. More generally, a state means everything that could in principle be known about the system, subject to the laws of physics. Quantum mechanics changes the form of that maximal description and makes the finite-volume counting discrete. Recovering the classical integral from the quantum sum is a genuine limiting argument, not a one-line substitution. [3]

1.6 A Heat Bath Made of Replicas

The hard question is now unavoidable: after equilibrium has been established, what determines P_i?

Place the subsystem in contact with a very large heat bath. Energy may pass back and forth, but the bath has so many degrees of freedom that the transfer of a small amount of energy barely changes its temperature. A particularly convenient bath consists of many weakly coupled replicas of the subsystem itself. One replica is the system under study; the others constitute its environment.

This construction often mirrors a real macroscopic system. Divide a large nearly homogeneous body into many similar cells. Each cell is a subsystem, and the remaining cells act as its bath. Denote the total number of replicas by N. The classroom demonstration uses nine replicas and three colored state labels, but the counting is independent of those particular numbers.

Let n_i be the number of replicas in state i. These are the *occupation numbers*. There may be infinitely many possible energy levels, but at finite total energy most very high levels have negligible or zero occupation.

The first constraint is simply the number of replicas:

$$\sum_i n_i = N. \tag{1.16}$$

The second is the total energy:

$$\sum_i n_i E_i = N\overline{E}. \tag{1.17}$$

The right-hand side is extensive. If the number of replicas is doubled while their statistical state is held fixed, the total energy doubles, whereas the mean energy per replica remains $\overline{E}$.

> **Question from the audience.** Must the replica construction also satisfy an energy constraint?
>
> **Response.** Yes. The occupation numbers must add to N, and their energy-weighted sum must equal the fixed total energy $N\overline{E}$. Those are precisely the two constraints that will later restrict the entropy maximum. [4]

[3] Lecture timestamp: 00:25:31–00:27:38.
[4] Lecture timestamp: 00:37:12–00:39:54.

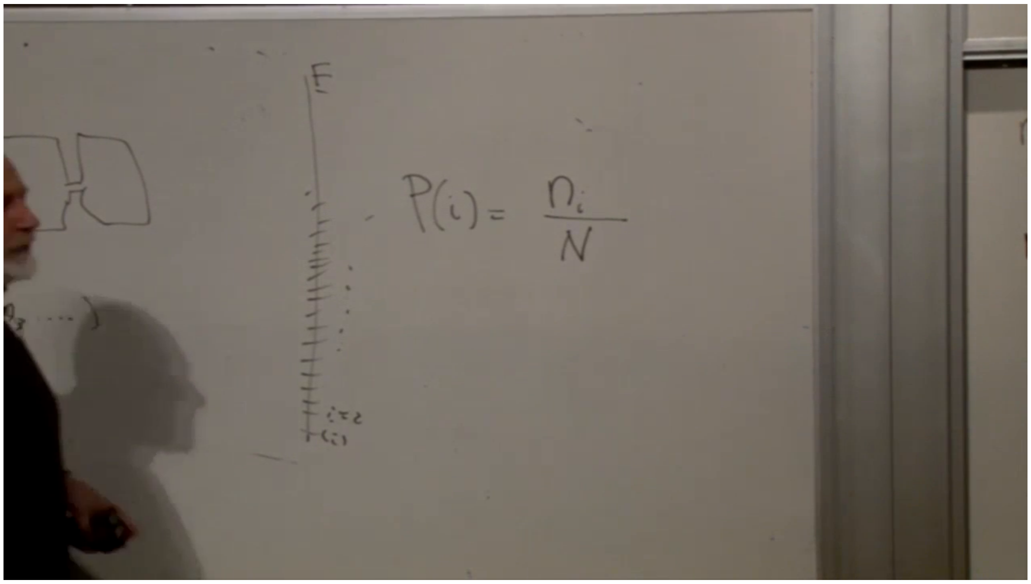

Figure 1.2: The energy-level picture and the identification of probability with the occupation fraction $P_i = n_i/N$.

Lecture frame: 00:41:40.

The probability that a randomly selected replica occupies state i is its relative frequency:

$$P_i = \frac{n_i}{N}, \qquad n_i = NP_i. \tag{1.18}$$

Question from the audience. Is the notation $N(i)$ on the board meant to differ from n_i?

Response. No. Both denote the occupation number of state i. We use n_i consistently and reserve capital N for the total number of replicas. [5]

Dividing (1.16) and (1.17) by N recovers the probability constraints:

$$\sum_i P_i = 1, \qquad \sum_i P_i E_i = \overline{E}. \tag{1.19}$$

The ensemble frequency interpretation has therefore reproduced the equations with which we began.

1.7 Multiplicity of an Occupation Pattern

Fix a set of occupation numbers $\{n_i\}$. How many assignments of the N distinguishable replicas to the states realize exactly that set? The answer is the multinomial coefficient

$$C(\{n_i\}) = \frac{N!}{\prod_i n_i!}. \tag{1.20}$$

To see the denominator, first arrange all N labeled replicas in $N!$ orders. Permuting the n_i replicas that occupy the same state does not produce a new occupation assignment, so divide by $n_i!$ for every state.

[5] Lecture timestamp: 00:41:28–00:41:33.

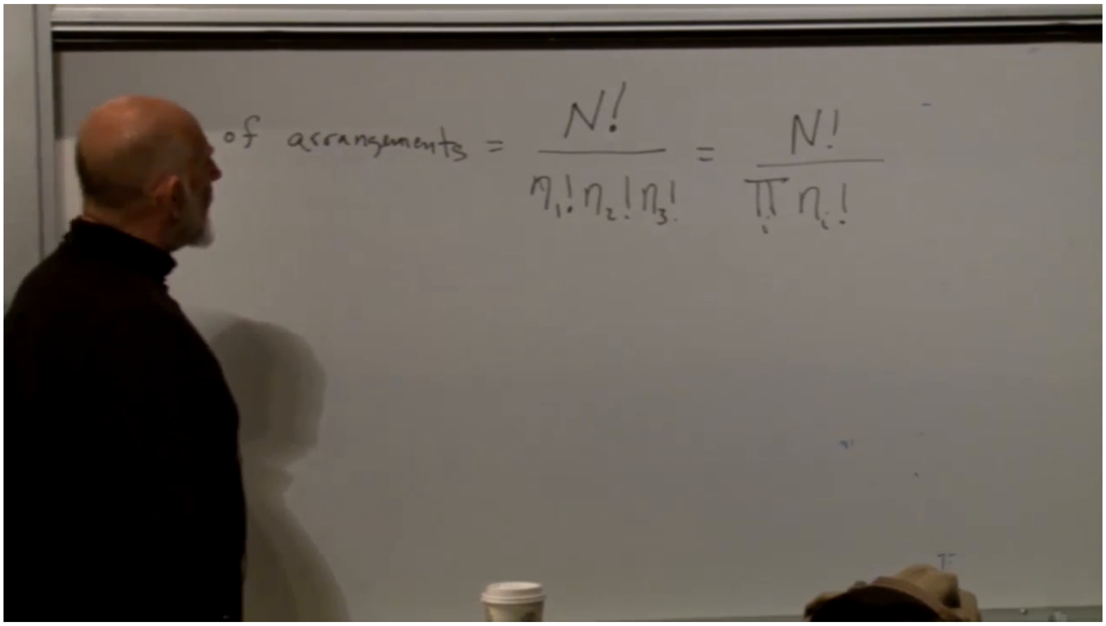

Figure 1.3: The multinomial multiplicity $C = N!/\prod_i n_i!$ written on the board.
Lecture frame: 00:46:55.

Unoccupied levels cause no problem because $0! = 1$. If all replicas occupy one state, then

$$C = \frac{N!}{N!} = 1,$$

as expected. If three replicas occupy three distinct specified states, then

$$C = \frac{3!}{1!\,1!\,1!} = 6.$$

With two replicas occupying two specified states, the formula gives two assignments: exchange which labeled replica occupies which state.

Question from the audience. Can more than one set of occupation numbers satisfy the two constraints?

Response. Usually many sets do. Equation (1.20) first counts the arrangements for one fixed set $\{n_i\}$, without yet deciding whether it satisfies the constraints. The next stage compares all constraint-compatible sets and finds which one has the largest multiplicity. [6]

The statistical assumption is that the microscopic assignments compatible with the isolated setup are treated symmetrically. A macroscopic occupation pattern realized by more assignments is therefore more probable. For large N, the distribution over the fractions n_i/N becomes sharply peaked. The equilibrium fractions are the values that maximize C, subject to (1.16) and (1.17).

[6] Lecture timestamp: 00:48:07–00:48:43.

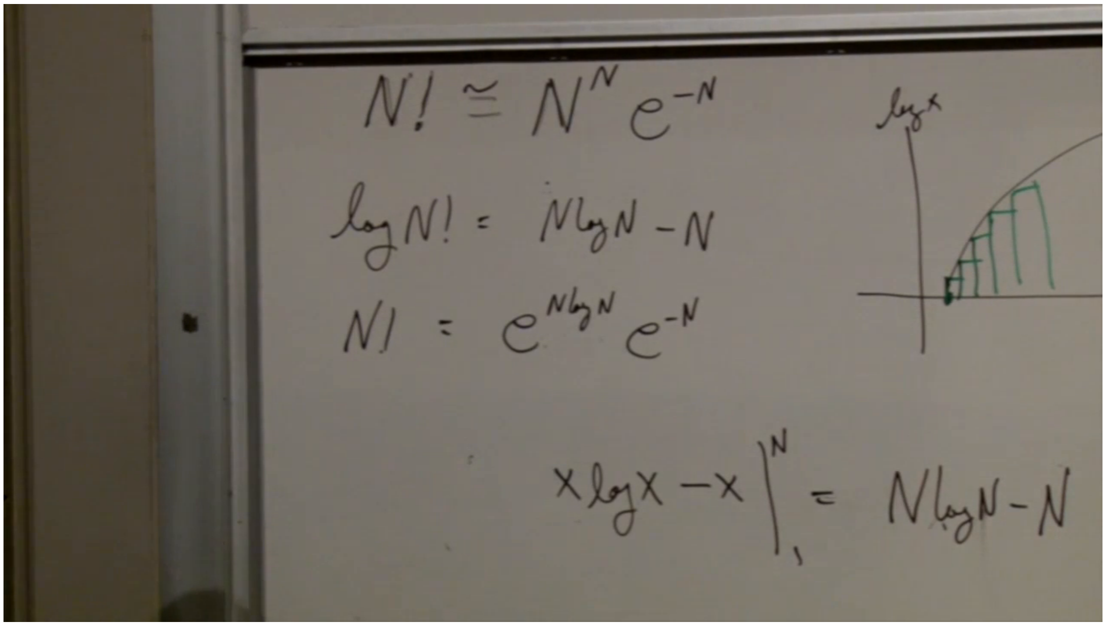

Figure 1.4: The leading Stirling approximation and its derivation by replacing the sum of $\log x$ with an integral.

Lecture frame: 01:03:45.

1.8 Stirling's Approximation

Factorials become enormous long before N reaches a macroscopic value, so we work with their logarithms. Begin with

$$\log N! = \sum_{x=1}^{N} \log x. \tag{1.21}$$

For large N, the sum is approximated by the area under $\log x$:

$$\begin{aligned} \log N! &\simeq \int_1^N \log x \, dx && (1.22)\\ &= [x \log x - x]_1^N && (1.23)\\ &= N \log N - N + 1. && (1.24) \end{aligned}$$

The antiderivative follows immediately from

$$\frac{d}{dx}(x \log x - x) = \log x.$$

At leading extensive order, the additive constant may be dropped:

$$\log N! \simeq N \log N - N. \tag{1.25}$$

Exponentiating gives the simplified form used in the lecture,

$$N! \simeq N^N e^{-N}. \tag{1.26}$$

Question from the audience. Does evaluating $x \log x - x$ at the lower limit leave an extra $+1$?

Response. Yes. Equation (1.24) contains that constant. It is negligible compared with the terms proportional to N as $N \to \infty$, which is why the leading formula omits it. The sum itself must begin at $x = 1$; $\log 0$ is not defined. [7]

Question from the audience. Is a factor proportional to $\sqrt{N}$ also missing?

Response. Yes. The sharper asymptotic formula is

$$N! \sim \sqrt{2\pi N} \left(\frac{N}{e}\right)^N .$$

Its logarithm adds only $\frac{1}{2}\log(2\pi N)$, which is subextensive compared with $N \log N - N$. It matters for accurate finite-N probabilities but not for the leading entropy calculation of a macroscopic system. [8]

Stirling's approximation must also be used with care in the denominator of (1.20). In the thermodynamic limit the states carrying nonzero limiting probabilities have $n_i = NP_i$, so their occupations grow with N. States with tiny or vanishing occupation can be handled by a finite-energy cutoff and then restored by a limit. The leading extensive entropy is unaffected by the omitted subleading terms.

1.9 The Logarithm of Multiplicity Is Entropy

Apply (1.26) to the numerator and all macroscopically occupied denominator factors:

$$C \simeq \frac{N^N e^{-N}}{\prod_i n_i^{n_i} e^{-n_i}}. \tag{1.27}$$

Since $\sum_i n_i = N$, the leading exponential factors cancel:

$$C \simeq \frac{N^N}{\prod_i n_i^{n_i}}. \tag{1.28}$$

It is easier to maximize a logarithm than a product, and the logarithm is monotone, so both maxima occur at the same occupation pattern:

$$\log C = N \log N - \sum_i n_i \log n_i. \tag{1.29}$$

Now substitute $n_i = NP_i$:

$$\log C = N \log N - \sum_i NP_i \log(NP_i) \tag{1.30}$$

$$= N \log N - N \sum_i P_i \log N - N \sum_i P_i \log P_i \tag{1.31}$$

$$= -N \sum_i P_i \log P_i, \tag{1.32}$$

[7] Lecture timestamp: 01:04:12–01:04:35.
[8] Lecture timestamp: 01:04:35–01:05:40.

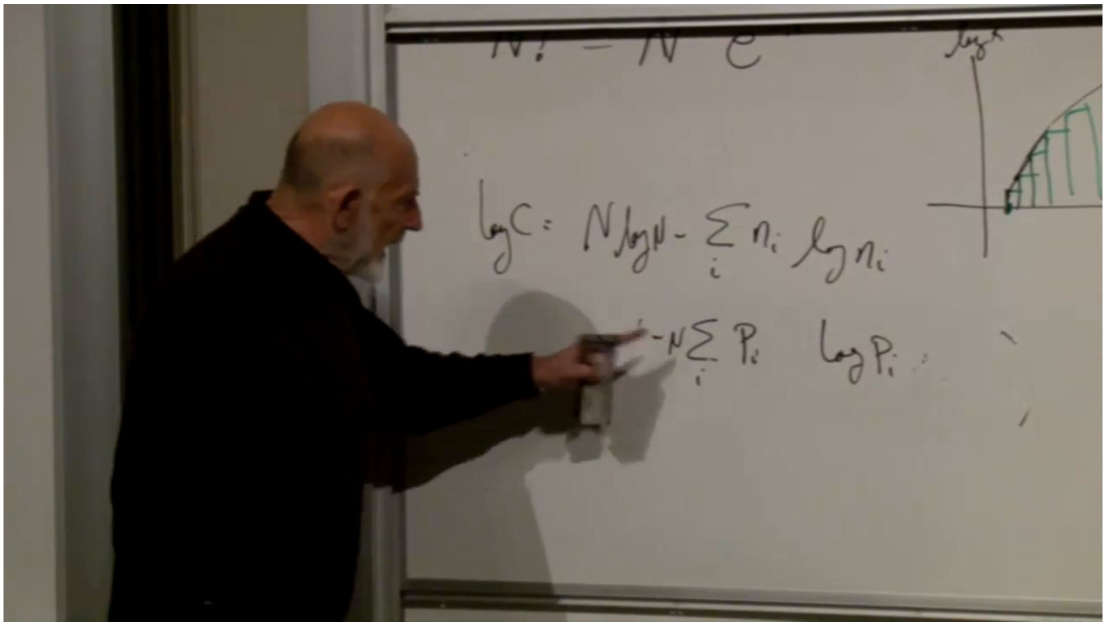

Figure 1.5: The board calculation connecting $\log C = N \log N - \sum_i n_i \log n_i$ to $-N \sum_i P_i \log P_i = NS$.

Lecture frame: 01:16:10.

where normalization cancels the two $N \log N$ terms. Thus, to leading order in N,

$$\boxed{\log C = NS}. \tag{1.33}$$

This relation is the conceptual center of the chapter. Entropy is not an unrelated quantity imposed on the counting problem. It is the multiplicity per replica, measured logarithmically. Maximizing the number of microscopic arrangements is therefore identical, at leading order, to maximizing the entropy of one replica.

Question from the audience. Would the multiplicity not be largest when all occupation numbers are equal?

Response. It is uniform when no state is distinguished and only $\sum_i n_i = N$ is imposed over a finite set of states. The fixed-energy constraint changes the answer because the values E_i are different. Equal occupations generally fail to produce the specified mean energy. [9]

Question from the audience. Capital N is large, but why may Stirling's approximation be applied to every n_i in the denominator?

Response. For each state with a nonzero limiting probability, $n_i = NP_i$ grows proportionally to N. One may first retain a finite energy range, take the large-N limit there, and then enlarge the range. Sparse tail states require separate finite corrections, but they do not alter the leading extensive entropy. [10]

[9] Lecture timestamp: 01:17:07–01:18:20.
[10] Lecture timestamp: 01:18:23–01:19:28.

Question from the audience. Is this large-N approximation what was meant earlier by coarse-graining?

Response. No. The thermodynamic limit controls large occupation numbers and asymptotic counting. Coarse-graining is the loss or deliberate discarding of microscopic resolution. They are distinct operations, even though both are often used in macroscopic statistical mechanics. [11]

Question from the audience. Since $\log C = N \log N - \sum_i n_i \log n_i$, should the second term be minimized rather than maximized?

Response. Yes; with N fixed, maximizing $\log C$ is equivalent to minimizing $\sum_i n_i \log n_i$. After substituting $n_i = NP_i$, the fixed $N \log N$ terms cancel and what remains is $-N \sum_i P_i \log P_i = NS$. Thus the same operation is described more naturally as maximizing the positive entropy S. [12]

Question from the audience. Why not finish the argument immediately by symmetry and set all the n_i equal?

Response. That shortcut ignores the energy labels. The normalization constraint treats the states symmetrically, but $\sum_i P_i E_i = \overline{E}$ does not: states of different energy contribute differently. The energy constraint is exactly where the nonuniform equilibrium distribution enters. [13]

1.10 The Maximum-Entropy Problem

We can now state the statistical-mechanical rule without ambiguity. Vary the probabilities P_i and maximize

$$S[P] = -\sum_i P_i \log P_i \tag{1.34}$$

subject to

$$\sum_i P_i = 1, \qquad \sum_i P_i E_i = \overline{E}. \tag{1.35}$$

The maximizing probabilities give the sharply preferred occupation fractions of a large replica ensemble. The occupation numbers fluctuate, but their relative deviations shrink as N grows, so the ratios n_i/N become well-defined thermodynamic probabilities.

Question from the audience. Are the E_i fixed state energies while the constraint uses a separately given total energy?

Response. Yes. The spectrum $\{E_i\}$ belongs to the subsystem. The complete replica collection has a fixed total energy, but energy continually moves among replicas. Dividing that total by N fixes the mean energy $\overline{E}$ of one replica, not its energy in each microscopic realization. [14]

The problem is easy to state and not yet solved. It is an extremization with two constraints, so one more mathematical tool is needed.

[11] Lecture timestamp: 01:19:29–01:19:46.
[12] Lecture timestamp: 01:19:48–01:20:55.
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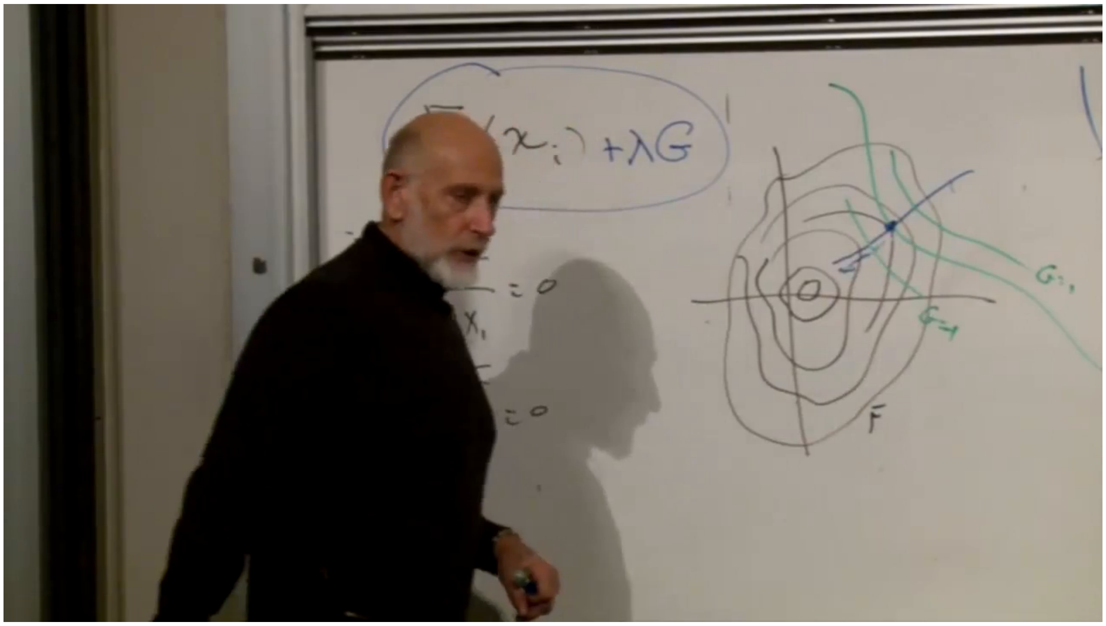

Figure 1.6: The board's contour geometry and the augmented function $F' = F + \lambda G$. A constrained extremum occurs at tangency.

Lecture frame: 01:37:10.

1.11 Lagrange Multipliers

For an unconstrained function $F(x_1, x_2, \ldots)$, a stationary point satisfies

$$\frac{\partial F}{\partial x_i} = 0 \qquad \text{for every } i. \tag{1.36}$$

Now require in addition that

$$G(x_1, x_2, \ldots) = 0. \tag{1.37}$$

> **Question from the audience.** What does x mean if the picture has coordinates x and y?
>
> **Response.** Here x is shorthand for the whole coordinate vector $(x_1, x_2, \ldots)$. In the two-dimensional drawing its components are simply x and y. Later the components will be the probabilities $P_1, P_2, \ldots$. [15]

Geometrically, the level curves $F =$ constant form a contour map, while $G = 0$ is the curve on which the system is allowed to move. A constrained extremum occurs where one contour of F is tangent to the constraint curve. At that point the gradients are parallel:

$$\nabla F = -\lambda \nabla G \tag{1.38}$$

for some number λ.

[15] Lecture timestamp: 01:30:12–01:30:20.

Equation (1.38) can be obtained by defining the augmented function

$$F' = F + \lambda G \tag{1.39}$$

and demanding ordinary stationarity:

$$\frac{\partial F'}{\partial x_i} = 0. \tag{1.40}$$

The multiplier λ is initially unknown. That does not leave the system underdetermined, because the original constraint $G = 0$ supplies one additional equation.

> **Question for the class.** Why does adding a term proportional to G make the constrained point stationary?
>
> **Response.** At the constrained extremum, motion tangent to $G = 0$ already produces no first-order change in F. The only remaining slope is normal to the constraint. Since ∇G points in that same normal direction, a suitable λ cancels it: $\nabla F + \lambda \nabla G = 0$. [16]

1.12 A Two-Variable Example

Minimize

$$F(x, y) = \frac{x^2 + y^2}{2} \tag{1.41}$$

subject to

$$x + y = 1. \tag{1.42}$$

Write the constraint as $G = x + y - 1 = 0$ and form

$$F' = \frac{x^2 + y^2}{2} + \lambda(x + y - 1). \tag{1.43}$$

The stationary equations are

$$\frac{\partial F'}{\partial x} = x + \lambda = 0, \tag{1.44}$$

$$\frac{\partial F'}{\partial y} = y + \lambda = 0. \tag{1.45}$$

Thus $x = y = -\lambda$. The remaining equation is the constraint,

$$x + y = -2\lambda = 1,$$

so

$$\lambda = -\frac{1}{2}, \qquad x = y = \frac{1}{2}. \tag{1.46}$$

The point $(1/2, 1/2)$ is where the smallest circle centered at the origin touches the line $x + y = 1$.

With several constraints, introduce one multiplier for each:

$$F' = F + \lambda_1 G_1 + \lambda_2 G_2 + \cdots . \tag{1.47}$$

[16] Lecture timestamp: 01:35:06–01:35:32.

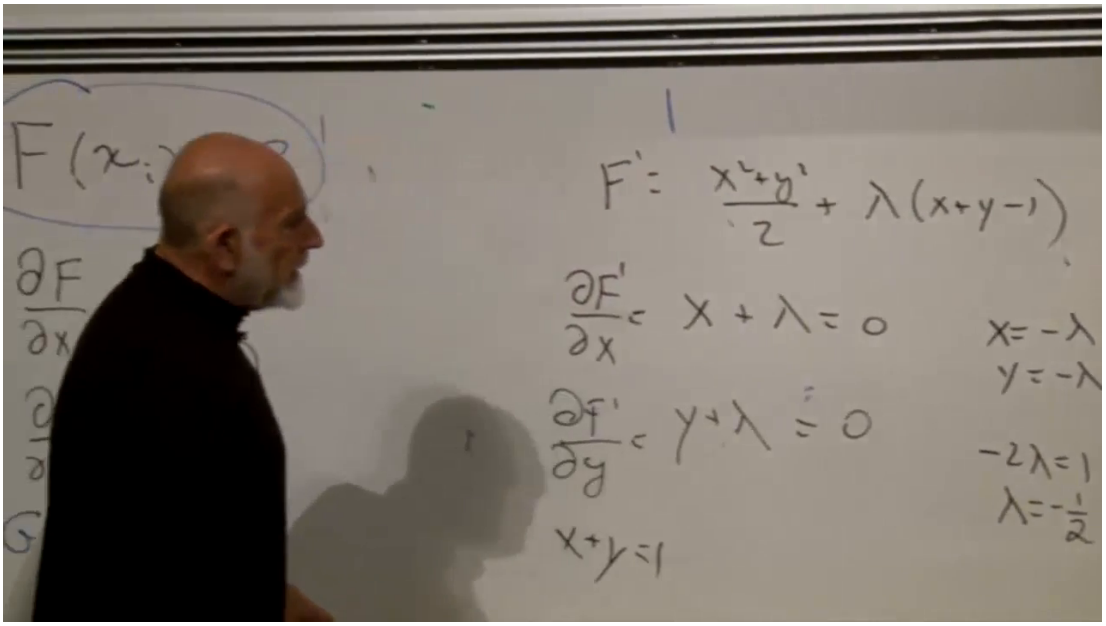

Figure 1.7: The completed Lagrange-multiplier calculation for minimizing $(x^2 + y^2)/2$ subject to $x + y = 1$.

Lecture frame: 01:44:10.

Then solve

$$\begin{aligned} \frac{\partial F'}{\partial x_i} &= 0 \quad \text{for every } i, \\ G_a &= 0 \quad \text{for every constraint } a. \end{aligned} \tag{1.48}$$

The lecture proposes a deliberately artificial practice problem: extremize a quadratic function of x, y, z while imposing two independent linear constraints, such as $x + y = 1$ and $z + x = 1$. Its purpose is not the particular answer but the rule that each constraint brings both one multiplier and one equation.

1.13 The Problem Ready for the Next Lecture

For statistical mechanics, the coordinates are the probabilities $\{P_i\}$, the function to be maximized is the entropy, and the constraints are normalization and fixed mean energy. Define

$$S[P] = -\sum_i P_i \log P_i, \tag{1.49}$$

$$G_1[P] = \sum_i P_i - 1, \tag{1.50}$$

$$G_2[P] = \sum_i P_i E_i - \overline{E}. \tag{1.51}$$

The augmented entropy may therefore be written

$$\Phi[P] = S[P] + \alpha G_1[P] + \beta G_2[P]. \tag{1.52}$$

The equations to solve are

$$\frac{\partial \Phi}{\partial P_i} = 0 \quad \text{for every } i, \qquad G_1 = 0, \qquad G_2 = 0. \tag{1.53}$$

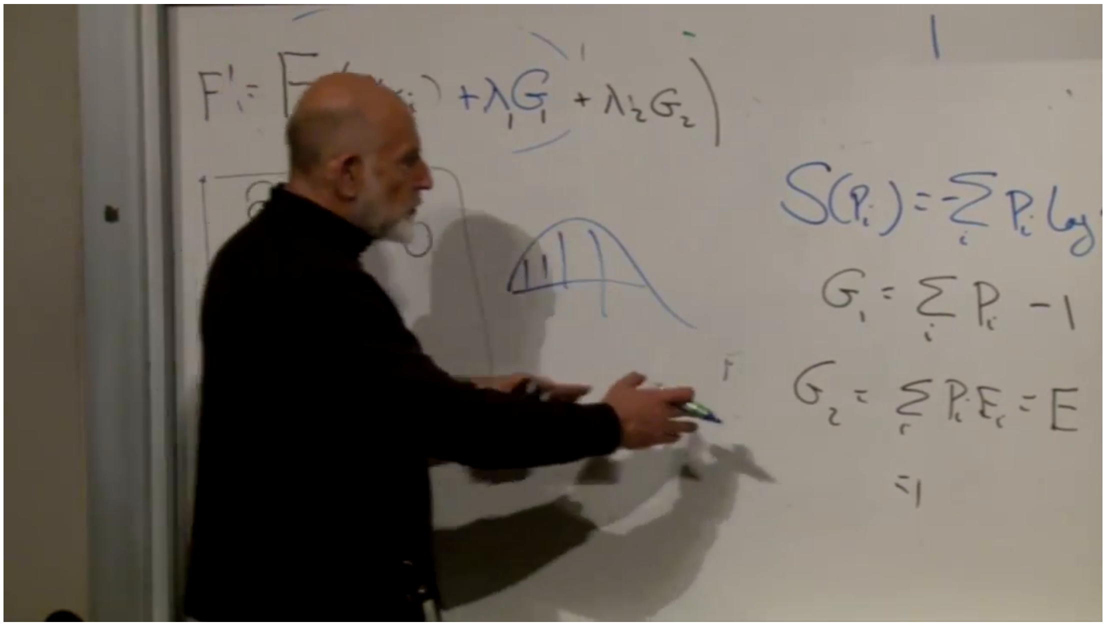

Figure 1.8: The closing board: entropy as a function of the probabilities, together with normalization and fixed mean energy.

Lecture frame: 01:53:05.

At this point the problem is completely posed. Solving (1.53) produces the equilibrium distribution. The multiplier conjugate to energy will become the inverse temperature $\beta = 1/T$, up to the sign convention chosen in (1.52). The next lecture carries out that differentiation and obtains the Boltzmann weights.